

Highly parallel genome-wide expression profiling of individual cells using nanoliter droplets

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Abstract

SummaryCells, the basic units of biological structure and function, vary broadly in type and state. Single-cell genomics can characterize cell identity and function, but limitations of ease and scale have prevented its broad application. Here we describe Drop-Seq, a strategy for quickly profiling thousands of individual cells by separating them into nanoliter-sized aqueous droplets, associating a different barcode with each cell's RNAs, and sequencing them all together. Drop-Seq analyzes mRNA transcripts from thousands of individual cells simultaneously while remembering transcripts' cell of origin. We analyzed transcriptomes from 44,808 mouse retinal cells and identified 39 transcriptionally distinct cell populations, creating a molecular atlas of gene expression for known retinal cell classes and novel candidate cell subtypes. Drop-Seq will accelerate biological discovery by enabling routine transcriptional profiling at single-cell resolution.

Introduction

Individual cells are the building blocks of tissues, organs, and organisms. Each tissue contains cells of many types, and cells of each type can switch among biological states. In most biological systems, our knowledge of cellular diversity is incomplete; for example, the cell-type complexity of the brain is unknown and widely debated (Luo et al., 2008; Petilla Interneuron Nomenclature et al., 2008). To understand how complex tissues work, it will be important to learn the functional capacities and responses of each cell type.

A major determinant of each cell's function is its transcriptional program. Recent advances now enable mRNA-seq analysis of individual cells (Tang et al., 2009). However, methods of preparing cells for profiling have been applicable in practice to just hundreds (Hashimshony et al., 2012; Picelli et al., 2013) or (with automation) a few thousand cells (Jaitin et al., 2014), typically after first separating the cells by flow sorting (Shalek et al., 2013) or microfluidics (Shalek et al., 2014) and then amplifying each cell's transcriptome separately. Fast, scalable approaches are needed to characterize complex tissues with many cell types and states, under diverse conditions and perturbations.

Here we describe Drop-Seq, a method to analyze mRNA expression in thousands of individual cells by encapsulating cells in tiny droplets for parallel analysis. Droplets – nanoliter-scale aqueous compartments formed by precisely combining aqueous and oil flows in a microfluidic device (Thorsen et al., 2001; Umbanhowar, 2000) – have been used as tiny reaction chambers for PCR (Hindson et al., 2011; Vogelstein and Kinzler, 1999) and reverse transcription (Beer et al., 2008). We sought here to use droplets to compartmentalize cells into nanoliter-sized reaction chambers for analysis of all of their RNAs. A basic challenge of using droplets for transcriptomics is to retain a molecular memory of the identity of the cell from which each mRNA transcript was isolated. To accomplish this, we developed a molecular barcoding strategy to remember the cell-of-origin of each mRNA. We critically evaluate Drop-Seq, then use it to profile cell states along the cell cycle. We then applied it to a complex neural tissue, mouse retina, and from 44,808 cell profiles retrieved 39 distinct populations, each corresponding to one or a group of closely related cell types. Our results demonstrate how large-scale single-cell analysis can help deepen our understanding of the biology of complex tissues and cell populations.

Results

Drop-Seq consists of the following steps (Figure 1A): (1) prepare a single-cell suspension from a tissue; (2) co-encapsulate each cell with a distinctly barcoded microparticle (bead) in a nanoliter-scale droplet; (3) lyse cells after they have been isolated in droplets; (4) capture a cell's mRNAs on its companion microparticle, forming STAMPs (Single-cell Transcriptomes Attached to Microparticles); (5) reverse-transcribe, amplify, and sequence thousands of STAMPs in one reaction; and (6) use the STAMP barcodes to infer each transcript's cell of origin.

A split-pool synthesis approach to generate large numbers of distinctly barcoded beads

To deliver large numbers of distinctly barcoded primer molecules into individual droplets, we use microparticles (beads). We synthesized oligonucleotide primers directly on beads (from 5' to 3', yielding free 3' ends available for enzymatic priming). Each oligonucleotide is composed of four parts (Figure 1B): (1) a constant sequence (identical on all primers and beads) for use as a priming site for downstream PCR and sequencing; (2) a "cell barcode" (identical across all the primers on the surface of any one bead, but different from the cell barcodes on other beads); (3) a Unique Molecular Identifier (UMI) (different on each primer, to identify PCR duplicates) (Kivioja et al., 2012); and (4) an oligo-dT sequence for capturing polyadenylated mRNAs and priming reverse transcription.

To efficiently generate massive numbers of beads, each with a distinct barcode, we developed a "split-and-pool" DNA synthesis strategy (Figure 1C). A pool of millions of microparticles is divided into four equally sized groups; a different DNA base (A, G, C, or T) is then added to each. All microparticles are then re-pooled, mixed, and re-split at random into another four groups, and then a different DNA base (A, G, C, or T) is added to each of the four new groups. After 12 cycles of split-and-pool DNA synthesis, the primers on any given microparticle possess the same one of $4^{12} = 16,777,216$ possible 12-bp barcodes, but different microparticles have different sequences (Figure 1C). The entire microparticle pool then undergoes eight rounds of degenerate oligonucleotide synthesis to generate the UMI on each oligo (Figure 1D); finally, an oligo-dT sequence (T30) is synthesized on the 3' end of all oligos on all beads.

To confirm that we could distinguish RNAs based on attached barcodes, we reverse-transcribed a pool of synthetic RNAs onto 11 microparticles and sequenced the resulting cDNAs (Figure S1A and Extended Experimental Procedures); 11 microparticle barcodes each constituted 3.5% – 14% of the resulting sequencing reads, whereas the next-most-abundant 12-mer constituted only 0.06% (Figure S1A). These results suggested that the microparticle-of-origin for most cDNAs can be recognized by sequencing. We also found that each bead contained more than 108 barcoded primer sites and that the sequence complexity of the barcodes approached theoretical limits (Figures S1B and S1C, Extended Experimental Procedures).

Microfluidics device for co-encapsulating cells with beads

We designed a microfluidic "co-flow" device (Utada et al., 2007) to co-encapsulate cells with barcoded microparticles (Figures 2A, S2 and DataFile 1). This device quickly co-flows two aqueous solutions across an oil channel to form more than 100,000 nanoliter-sized droplets per minute. One flow contains the barcoded microparticles suspended in a lysis buffer; the other flow contains a cell suspension (Figure 2A, left, Figure 2B). The number of droplets created greatly exceeds the number of beads or cells injected, so that a droplet will generally contain zero or one cells, and zero or one beads. Millions of nanoliter-sized droplets are generated per hour, of which thousands contain both a bead and a cell (Movie S1). STAMPs are produced in the subset of droplets that contain both a bead and a cell.

Sequencing and analysis of many STAMPs in a single reaction

To efficiently process thousands of STAMPs at once, we break droplets, collect the mRNA-bound microparticles, and reverse-transcribe the mRNAs (from the microparticle-attached primers) together in one reaction, forming covalent, stable STAMPs (Figure 2A, step 7, and Experimental Procedures). A scientist can then select any desired number of STAMPs for the preparation of 3'-end digital expression libraries (Figure 2C, Experimental Procedures). We sequence the resulting molecules from each end (Figure 2C) using high-capacity parallel sequencing. We digitally count the number of mRNA transcripts of each gene ascertained in each cell, using the UMIs to avoid double-counting sequence reads that arose from the same mRNA transcript. We thereby create a matrix of digital gene-expression measurements (one measurement per gene per cell) for further analysis (Figure 2D, Experimental Procedures).

The single cell accuracy and sensitivity of Drop-Seq libraries

To measure the accuracy with which Drop-Seq remembers the cell-of-origin of each mRNA, we analyzed mixtures of cultured human (HEK) and mouse (3T3) cells, scoring the numbers of human and mouse transcripts that associated with each cell barcode (Figure 3A, 3B, S3A). We found that the individual STAMPs created by Drop-Seq were highly organism-specific (Figure 3A, 3B), indicating high single-cell integrity of the libraries. At saturating levels of sequence coverage, we detected an average of 44,295 mRNA transcripts from 6,722 genes in HEK cells and 26,044 transcripts from 5,663 genes in 3T3 cells (Figures 3C and 3D).

To understand how Drop-Seq libraries compare to other single-cell methods, we used three quality metrics: (i) the frequency of cell-cell doublets; (ii) single-cell purity; and (iii) transcript capture rates.

Cell states: Drop-Seq analysis of the cell cycle

To evaluate the visibility of cell states in Drop-Seq, we first examined cell-to-cell variation among the 589 HEK and 412 3T3 STAMPs shown in Figure 3B. Both cultures consisted of asynchronously dividing cells; principal components analysis

(PCA) of the single-cell expression profiles showed the top principal components to be dominated by genes with roles in protein synthesis, growth, DNA replication, and other aspects of the cell cycle. We inferred the cell-cycle phase of each of the 1,001 cells by scoring for gene sets (signatures) reflecting five phases of the cell cycle previously characterized in chemically synchronized cells (G1/S, S, G2/M, M, and M/G1) (Figure 4A, Table S2) (Whitfield et al., 2002). We identified 544 human and 668 mouse genes with expression patterns that varied along the cell cycle (at a false discovery rate of 5%; Experimental Procedures) (Figure 4B), including 200 orthologous gene pairs ($p < 10^{-65}$ by hypergeometric test). Of these orthologous gene pairs, most (82.5%) have been previously annotated as related to the cell cycle in at least one species; among the other 17.5%, we found some that would be expected to show cell cycle variation (e.g. E2F7 and PARPBP) and many that to our knowledge were not previously connected to the cell cycle (Figure 4C and Table S3). Single-cell analysis at this scale enabled characterization of cell-cycle gene expression without chemical synchronization and at high temporal resolution.

Cell types: Drop-Seq analysis of the retina

We selected the retina as the first tissue to study with Drop-Seq because decades of work has generated molecular information about many retinal cell types (Masland, 2012; Sanes and Zipursky, 2010), allowing us to relate our RNA-seq data to prior classification. The retina contains five neuronal classes—retinal ganglion, bipolar, horizontal, photoreceptor, and amacrine—each defined by morphological, physiological, and molecular criteria (Figure 5A). Most of the classes are divisible into discrete types – a total currently estimated at about 100 – but well under half of these types possess known, distinguishing molecular markers.

We sequenced 49,300 STAMPs prepared from 14-day-old mouse retinas (STAMPs were collected in seven batches over four days). We performed principal components analysis on the 13,155 largest libraries (Figure S5, Table S3), then reduced the 32 statistically significant PCs (Experimental Procedures) to two dimensions using t-Distributed Stochastic Neighbor Embedding (tSNE) (Amir el et al., 2013; van der Maaten and Hinton, 2008). We projected the remaining 36,145 cells in the data into the tSNE analysis. We then combined a density clustering approach with post hoc differential expression analysis to divide 44,808 cells among 39 transcriptionally distinct clusters (Extended Experimental Procedures) ranging from 50 to 29,400 cells (Figures 5B and 5C). Finally, we organized the 39 cell populations into larger categories (classes) by building a dendrogram of similarity relationships among the 39 cell populations (Figure 5D, left).

The cell populations inferred from this analysis were readily matched to the known retinal cell types, including all five neuronal cell classes, based on the specific expression of known markers for these cell types (Figure 5D, right, and Figure S6A). Additional clusters corresponded to astrocytes (associated with retinal ganglion cell axons exiting the retina), resident microglia, endothelial cells (from intra-retinal vasculature), pericytes, and fibroblasts (Figure 5D). The relative abundances of the major cell classes in our data agreed with earlier estimates from microscopy (Jeon et al., 1998) (Table 1).

Replication and cumulative power of Drop-Seq data

Replication across experimental sessions enables the construction of cumulatively powerful datasets – but only if data are replicable and comparable. The retinal STAMPs were generated on four different days (weeks apart), utilizing different litters and multiple runs in several sessions, for a total of seven replicates. One of the runs was performed at a particularly low cell concentration (15 cells/ μ L) and thus high purity, to evaluate whether results were artifacts of cell-cell doublets or single-cell impurity. We found that all 39 clusters contained cells from every experiment. One cluster (arrow in Figure 5E; star in Figure S6B), which drew disproportionately from two replicates, expressed markers of fibroblasts, a nonretinal cell type that is present in tissue surrounding the retina, and hence likely represents imprecise dissection.

We examined how the classification of cells (based on their patterns of gene expression) evolved as a function of the numbers of cells in analysis. We used 500, 2,000, or 9,731 cells from our dataset, and asked how (for example) cells identified as amacrine in the full dataset clustered in analyses of smaller numbers of cells (Figure 5F). As the number of cells in the data increased, distinctions between related clusters become clearer, stronger, and finer in resolution, with the result that a greater number of rare amacrine cell sub-populations (each representing 0.1–0.9% of the cells in the experiment) could ultimately be distinguished from one another (Figure 5F).

Profiles of amacrine cell types

To characterize distinctions among closely related cell populations, we focused on the 21 clusters of amacrine. Amacrine are the most morphologically diverse neuronal class (Masland, 2012), but the majority of types lack defining molecular markers. Most amacrine cells are inhibitory, utilizing either GABA or glycine as a neurotransmitter. Excitatory amacrine cells that release glutamate have also been identified (Haverkamp and Wässle, 2004). Another amacrine cell population expresses no GABAergic, glycinergic or glutamatergic markers; its neurotransmitter is unidentified (nGnG

amacrines) (Kay et al., 2011).

We first identified markers that were most universally expressed by amacrines relative to other cell classes (Figure 6A). We then assessed the expression of known glycinergic and GABAergic markers; their mutually exclusive expression is a fundamental distinction among amacrines. Of the 21 amacrine clusters, 12 were identifiable as GABAergic (Gad1 and/or Gad2-positive) and 5 others were glycinergic (glycine transporter Slc6a9-positive) (Figure 6B). An additional cell population was identified as excitatory by its expression of a glutamate transporter, Slc17a8 (Figure 6B). The remaining three clusters (clusters 4, 20, and 21) had low levels of GABAergic, glycinergic, and glutamatergic markers; these likely include nGnG amacrines.

Among the glycinergic and GABAergic clusters, we found many amacrine types with known markers. A-II amacrine neurons appeared to correspond to the most divergent glycinergic cluster (Figure 6B, cluster 16), as this was the only cluster to strongly express the Gjd2 gene encoding the gap junction protein connexin 36 (Feigenspan et al., 2001). Ebf3, a transcription factor found in SEG glycinergic as well as nGnG amacrines, was specific to clusters 17 and 20. Starburst amacrine neurons (SACs), the only retinal cells that use acetylcholine as a co-transmitter, were identifiable as cluster 3 by their expression of the cholinergic marker Chat (Figure 6B). Unlike other GABAergic cells, SACs expressed Gad1 but not Gad2, as previously observed in rabbit (Famiglietti and Sundquist, 2010).

We then identified selectively expressed markers for each of the 21 amacrine cell populations (Figure 6C and Table S4). We validated two of the markers immunohistochemically. First, we co-stained retinal sections with antibodies to the transcription factor MAF, the top marker of cluster 7, plus antibodies to either GAD1 or SLC6A9, markers of GABAergic and glycinergic transmission, respectively. As predicted by the Drop-Seq analysis, MAF was found in a small subset of amacrine cells that were GABAergic and not glycinergic (Figure 6D). Cluster 7 had numerous genes that were enriched relative to its nearest neighbor, cluster 6 (Figure 6E, 16 genes > 2.8-fold enrichment, $p < 10^{-9}$), including Crybb3, which belongs to the crystallin family of proteins that are known to be directly upregulated by Maf (Yang and Cvekl, 2005), and another, the protease Mmp9, which accepts crystallins as substrates (Descamps et al., 2005). Second, we stained sections with antibodies to PPP1R17 (Figure 6F). Cluster 20 shows weak, infrequent glycine transporter expression and is one of only two clusters (with cluster 21) that express Neurod6, a marker of nGnG neurons (Kay et al., 2011). We used a transgenic strain (MitoP) that has been shown to express cyan fluorescent protein (CFP) specifically in nGnG amacrines (Kay et al., 2011). PPP1R17 stained 85% of all CFP-positive amacrines in the MitoP line, validating this as a marker of nGnG cells (Figure 6F). PPP1R17 was one of several markers that distinguished Cluster 20 from its closest neighbor, Cluster 21 (Figure 6G; 12 genes > 2.8-fold enrichment, $p < 10^{-9}$). The differences between Clusters 20 and 21 suggest a hitherto unsuspected level of heterogeneity among nGnG amacrines.

Supervised analysis reveals additional diversity

Our unsupervised analysis grouped cells into 39 transcriptionally distinct populations, but morphological and functional criteria suggest that there are ~100 retinal cell types. We asked whether supervised analysis could reveal multiple types within individual clusters. For example, retinal ganglion cells (RGCs), which consist of about 30 types (Sanes and Masland, 2015), formed a single cluster in our analysis, perhaps because it is a rare cell population (1%, Table 1). Five RGC types, called intrinsically photosensitive RGCs (ipRGCs), express Opn4, the gene encoding the photopigment melanopsin. Opn4⁺ RGCs (26/432) expressed nine genes at levels two-fold higher than Opn4⁻ RGCs ($p < 10^{-9}$, Figure 6H), including Tbr2/Eomes, known to be a selective marker for this population (Sweeney et al., 2014). This result reveals additional heterogeneity that may also emerge *ab initio* as analyses expand to include more cells.

Discussion

Ascertaining transcriptional variation across individual cells is a valuable way of learning about complex tissues and functional responses, but single-cell analysis has been limited by the time and cost of preparing libraries from many individual cells. A scientist employing Drop-Seq can prepare 10,000 single-cell libraries for sequencing in twelve hours, for about 6.5 cents per cell (Table S5), representing a >100-fold improvement in both time and cost relative to existing methods. A Drop-Seq setup can be constructed quickly and inexpensively in a standard biology lab using readily available equipment (Figure S2B and Extended Experimental Procedures). We hope that ease, speed, and low cost facilitate exuberant experimentation, careful replication, and many cycles of experiments, analyses, ideas, and more experiments.

In validating Drop-Seq, we developed stringent species-mixing experiments to measure single-cell purity and cell doublet rates in our libraries. In another article in this issue, Klein et al. (Klein, 2015) describe a droplet-based approach to single-cell RNA-seq, and also use species-mixing experiments to evaluate it. Our results indicate that all methods of isolating single cells from a cell suspension, including Drop-Seq, fluorescence activated cell sorting (FACS) and microfluidics, are vulnerable to impurities, and highlight the value of performing species mixing experiments to assess

single-cell approaches. In our retina analysis, even relatively impure libraries generated in “ultra-high-throughput” modes (100 cells per μL , allowing the processing of 10,000 cells per hour at ~10% doublet and impurity rates) appeared to yield a robust and biologically validated cell classification, but other tissues or applications may require using Drop-Seq in purer modes.

Unsupervised computational analysis of Drop-Seq data identified 39 transcriptionally distinct retinal cell populations, many representing specific subtypes of the major retinal cell classes (Figures 5 and 6). It is a particular strength of the retina that establishing correspondence between cluster and type was in many cases straightforward; an important direction will be to identify cell types and states in other parts of the brain—as well as in other tissues—about which less is currently known.

We see many applications of Drop-Seq, beyond the identification of cell types and cell states. Genome-scale genetic studies are identifying many genes whose variation contributes to disease risk, but biology has lacked similarly high-throughput ways of connecting these genes to specific cell populations and unique functional responses. Drop-Seq could be used to provide initial insights into how these genes function in the diverse cell types composing each tissue. In addition, coupling Drop-Seq to perturbations — such as small molecules, mutations, pathogens, or other stimuli —could generate an information-rich, multi-dimensional readout of the influence of perturbations on many kinds of cells.

The functional implications of a gene's expression are a product not just of that gene's intrinsic properties, but also of the entire cell-level context in which the gene is expressed. We hope Drop-Seq enables the abundant and routine discovery of such relationships in many areas of biology.

Experimental Procedures

Device design and fabrication

Microfluidic devices were designed using AutoCAD software (Autodesk, Inc.), and the components tested using COMSOL Multiphysics (COMSOL Inc.). Full details are described in Extended Experimental Procedures.

Barcoded microparticle synthesis

Bead functionalization and reverse-direction phosphoramidite synthesis were performed by Chemgenes Corp (Wilmington, MA). “Split-and-pool” cycles were accomplished by removing the dry resin from each column, hand mixing, and weighing out four equal portions before returning the resin for an additional cycle of synthesis. Full details are described in Extended Experimental Procedures.

Drop-Seq procedure

Monodisperse droplets ~1 nL in size were generated using the microfluidic device described in Extended Experimental Procedures, in which barcoded microparticles, suspended in lysis buffer, were flowed at a rate equal to that of a single-cell suspension, so that resulting droplets were composed of an equal amount of each component. As soon as droplet generation was complete, droplets were broken with perfluorooctanol in 30 mL of 6x SSC. The addition of a large aqueous volume to the droplets reduces hybridization events after droplet breakage, because DNA base pairing follows second-order kinetics (Britten and Kohne, 1968; Wetmur and Davidson, 1968). The beads were then washed and resuspended in a reverse transcriptase mix, followed by a treatment with exonuclease I to remove unextended primers. The beads were then washed, counted, aliquoted into PCR tubes, and PCR amplified. The PCR reactions were purified and pooled, and the amplified cDNA quantified on a BioAnalyzer High Sensitivity Chip (Agilent). The cDNA was fragmented and amplified for sequencing with the Nextera XT DNA sample prep kit (Illumina) using custom primers that enabled the specific amplification of only the 3' ends (Table S6). The libraries were purified, quantified, and then sequenced on the Illumina NextSeq 500. All details regarding reaction conditions, primers used, and sequencing specifications can be found in the Extended Experimental Procedures.

Cell cycle analysis of HEK and 3T3 cells

Gene sets reflecting five phases of the HeLa cell cycle (G1/S, S, G2/M, M and M/G1) were taken from Whitfield et al. (Whitfield et al., 2002) with some modification (Extended Experimental Procedures and Table S2). A phase-specific score was generated for each cell, across all five phases, using averaged normalized expression levels ($\log_2(\text{TPM}+1)$) of the genes in each set. Cells were then ordered along the cell cycle by comparing the patterns of these five phase scores per cell. To identify cell cycle-regulated genes, we used a sliding window approach, and identified windows of maximal and minimal average expression, both for ordered cells, and for shuffled cells, to evaluate the false-discovery rate. Full details may be found in Extended Experimental Procedures.

Principal components and clustering analysis of retina data

The clustering algorithm for the retinal cell data was implemented and performed using Seurat, a recently developed R package for single-cell analysis (Satija et al., 2015). Principal components analysis (PCA) was first performed on a 13,155-cell “training set” of the 49,300-cell dataset, using single-cell libraries in which transcripts from > 900 genes were detected. We found this approach was more effective in discovering structures corresponding to rare cell types than performing PCA on the full dataset, which was dominated by numerous, tiny rod photoreceptors (Extended Experimental Procedures). Thirty-two statistically significant PCs were identified using a permutation test and independently confirmed using a modified resampling procedure (Chung and Storey, 2014). We projected individual cells within the training set based on their PC scores onto a single two-dimensional map using t-Distributed Stochastic Neighbor Embedding (t-SNE) (van der Maaten and Hinton, 2008). The remaining 36,145 single-cell libraries (< 900 genes detected) were next projected on this t-SNE map, based on their representation within the PC-subspace of the training set (Berman et al., 2014; Shekhar et al., 2014). This approach mitigates the impact of noisy variation in the lower complexity libraries due to gene dropouts. It was also reliable in the sense that when we withheld from the t-SNE all cells from a given cluster and then tried to project them, these withheld cells were not spuriously assigned to another cluster by the projection (Table S7). Point clouds on the t-SNE map represent candidate cell types; density clustering (Ester et al., 1996) identified these regions. Differential expression testing (McDavid et al., 2013) was then used to confirm that clusters were distinct from each other. Hierarchical clustering based on Euclidean distance and complete linkage was used to build a tree relating the clusters. We noted expression of several rod-specific genes, such as *Rho* and *Nrl*, in every cell cluster, an observation that has been made in another retinal cell gene expression study (Siegert et al., 2012) and likely arises from solubilization of these high-abundance transcripts during cell suspension preparation. Additional information regarding retinal cell data analysis can be found in the Extended Experimental Procedures.

Data availability

Both raw and analyzed data have been deposited at Gene Expression Omnibus Accession GSE63473.

Supplementary Material:

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